

Transmission Media

Module 22

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my dear students hello I welcome you all in our series of lectures on cyber security today we are going to study about transmission media we will also learn how to measure the performance of various transmission Medias so let us begin with it transmission media is a pathway that carries the information from sender to receiver we use different types of cables to transmit data data is transmitted normally through electrical or electromagnetic signals an electrical signal is in the form of current an electromagnetic signal is series of electromagnetic energy pulses at various frequencies these signals can be transmitted through copper wires Optical fibers atmosphere water and vacuum different medias have different properties like bandwidth delay cost and ease of installation and maintenance transmission media is also called

Communication channel the figure shows the position of transmission media in the relation to a physical layer now what are the various factors for design of data transmission systems a key concern generally is data rate and distance the the greater the data rate and distance the better a number of design factors relating to the transmission medium and to the signal determine the data rate and the distance such as number one band width all other factors remaining constant the greater the bandwidth of a signal the higher the data rate that can be achieved number two transmission impairment impairments such as attenuation limit the distance for guided media Twisted paer generally suffer more impairment than coxl cable

which in turn suffers more than optical fiber number three interference interference from competing signals in overlapping frequency bands can dist or wipe out a signal interference is of particular concern for unguided media but it is also a problem with guided media proper shielding of a guided medium can minimize this problem number four number of receivers a guided medium can be used to construct a point to-point link or a sheer link with multiple attachments in the later case each attachment introduces some attenuation and distortion on the line limited distance and the data rate next is types of transmission media the transmission media is categorized as number one gu guided media for guided transmission media the transmission

capacity in terms of either data rate or bandwidth depends critically on the distance and on whether the medium is point to point or multi-point the three guided media commonly used for data transmission are twisted pair coxl cable and optical fiber let us now learn about twisted pair a twisted pair consists of two conductors normally copper each with its own plastic insulation one of the wires is used to carry signals to the receiver and the other is used only as a ground reference the receiver uses the difference between the two and addition to the signal sent by the sender on one of the wires interference and cross talk may affect both wires and create unwanted signals if the two wires are parallel the effect of these unwanted

signal is not the same in both wires because they are at different locations relative to the noise or cross talk sources this results in a difference at the receiver by twisting the pairs a balance is maintained for example suppose in one twist one wire is closer to the noise source and the other is farther in the next twist the reverse is true twisting makes it probable that both wires are equally affected by external influences such as noise or cross talk this means that the receiver which calculates the difference between the two receives no unwanted signals The Unwanted signals are mostly canceled out the number of twist per unit of length called has some effect on the quality of the cable twisted pair is of two types unshielded twisted pair cable and shielded twisted pair

cable the most common twisted pair cable used in Communications is referred to as unshielded Twisted paer cable shielded Twisted paer cable has a metal foil or braided mesh covering that encases each pair of insulated conductors although metal casing improves the quality of the cable by preventing the penetration of noise or cross talk it is bulkier and more expensive although UTP was once considered to be slower at transmitting data than other types of cable this is no longer true in fact UTP is considered the fastest copper based medium today the main features of unshielded Twisted cable are number one speed and throughput is from 10 to th000 MVPs number two average cost per node is very less and number three media and connector size used with it is very

small next is connectors the most common UTP connector is RJ45 where RJ stands for registered Jack then comes shielder twisted pair cable shielder twisted pair cable combines the techniques of shielding cancellation and wire twisting each pair of wires is wrapped in a metallic foil the four pairs of wires then are wrapped in an overall metallic braid or foil usually 150 ohm cable as specified for use in ether Network installations shielder Twisted paer reduces electrical noise both within the cable and from outside the cable Shield twisted pair usually is installed with STP data connector which is created especially for the STP cable however STP cabling also can use the same RG connectors that UTP

uses although STP prevents interference better than UTP it is more expensive and difficult to install in addition the metallic shielding must be grounded at both ends if it is improperly grounded The Shield acts like an antenna and picks up unwanted signals because of its cost and difficulty with termination STP is rarely used in Ethernet networks STP is primly used in

Europe the main features of STP cable are number one speed and throughput is from 10 to 100000 Mbps number two average cost per node is moderately High number three media and connector size used with it varies from medium to large next is applications of Twisted pairs there are many applications such as UTP cables are found in many ethernet

networks UTP cables are used in telephone systems it is the most commonly used medium for Workhorse for communications within buildings in the telephone system individual residential telephone lines are connected to the local telephone exchange or end office by twisted pair wire these are referred to as subscriber Loops there are many advantages associated with twisted pair cables such as their installation is easy they're flexible and cheap and they have a high speed capacity the disadvantage is that their bandwidth is low when compared with coaxial cable and they provide less protection from interference next is coaxial cables coaxial is called by this name because it contains two conductors that are parallel to each other copper is used in this a center

conductor which can be a solid wire or a stranded one it is surrounded by PVC insulation 9 minutes, 1 second a sheath which is encased in an outer conductor of metal foil outer metallic wrapping is used as a shield against noise and the second conductor which completes the circuit the outer conductor is also encased in an insulating sheet the outermost part is the plastic cover which protects the whole cable the most common coaxial standards are number one 50 ohm RG7 or RG11 which is used with thick ethernet number two 50 ohm RG58 used with thin ethernet number three 75 ohm RG59 which is used with cable television and number four 93 ohm RG62 used with ethernet the figure shows the coaxial cable next is coaxial cable connectors to connect coaxial cable to devices we need coaxial

connectors the most common type of connector used today is the BNC connector the BNC T connector and the BNC Terminator the BNC connector is used to connect the end of the cable to a device such as a TV set the BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device the BNC Terminator is used at the end of the cable to prevent the reflection of the signal now what about the performance of the coaxial cables the attenuation is much higher in coaxial cables than in the twisted pair cables although coaxial cable has a much higher bandwidth the signal weakens rapidly and requires the frequent use of repeaters next is various applications of coaxial cables number one coaxial cable was widely used in analog telephone networks

where a single coaxial Network could carry 10,000 voice signals number two it was used in digital telephone networks where a single coaxial cable could carry Digital Data up to 600 Mbps however coaxial Cable in telephone networks has largely been replaced today with fiber optic cables number three cable TV networks also use coaxial cables fourth because of its high bandwidth and consequently High data rate coaxial cable was chosen for digital Transmission in early ethernet networks next is various types of coaxial cables there are two types of coaxial cables baseband and Broadband baseband coaxial cable is a 50 ohm coaxial cable which is used for digital

transmission it is mostly used for less. Baseband transmits a single signal at a time with very high speed the major drawback is that it needs amplification.

after every 1000 ft. Broadband coaxial cable uses analog transmission on standard cable television cabling it transmits several simultaneous signals using different frequencies it covers large area when compared with baseband coaxial cables it has many advantages such as:
 number one bandwidth is high
 number two used in long distance telephone lines
 number three transmits digital signals at a very high rate of 10 Mbps
 number four has much higher noise immunity
 number five performs data transmission without distortion and
 number six they can span to longer distance at higher speeds as they have better shielding when compared to Twisted pair cables it has some disadvantages also such as:
 number one single cable failure can fail the entire network
 number two difficult to install

and expensive when compared with twisted pair and
 number three if the shield is imperfect it can lead to grounded loop.
 next is fiber optic cable a fiber optic cable is made of glass or plastic and transmit signals in the form of light. Figure shows the optical fiber light travels in a straight line as long as it is moving through a single uniform substance if a ray of light traveling through one substance suddenly enters another substance of a different density the ray changes direction as the figure shows if the angle of incidence I which is the angle the Ray makes with the line perpendicular to the interface between the two substances is less than the critical angle the ray reflects and moves closer to the

surface if the angle of incidence is equal to the critical angle the light bends along the interface if the angle is greater than the critical angle the ray reflects back and travels again in the denser substance note that the critical angle is a property of the substance and its value differs from one substance to another Optical fibers use refraction to guide light through a channel figure shows the bending of a light Ray fiber optic cable has a bandwidth of more than 2 GB per second next is propagation modes current technology supports two modes called multi mode and single mode for propagating light along Optical channels each requiring fiber with different physical characteristics

multi mode can be implemented in two forms step index or graded index multiple beams from a light source move through the core in different paths how these beams move within the cable depends on the structure of the core in multi mode step index fiber the density of the core remains constant from the center to the edges a beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding at the interface there is an abrupt change due to a lower density this alters the angle of the beam's motion the term step index refers to the suddenness of this change which contributes to the distortion of the signal as it passes through the fiber a second type of fiber called multi mode graded index fiber decreases

this Distortion of the signal through the C the word index here refers to the index of refraction the index of refraction is related to density density is highest at the center of the core and decreases gradually to its lowest at the edge single mode uses step index fiber and a highly focused source of FL that limits beams to a small range of angles all close to the horizontal figure shows various fiber optic transmission modes next is fiber optic cable connectors there are three types of connectors for fiber optic cables the subscriber Channel connector is used for cable TV it uses a push pull Locking System the straight type connector is used for connecting cable to network devices it uses a Bonet

Locking System and is more liable than SC Mt RJ is another connector that is used and is the same size as RJ45 next comes applications of fiber optic cables first application is fiber optic cable is often found in backbone networks because it's wide bandwidth is cost effective number two some cable TV companies use a combination of optical fiber and coxl cable thus creating a hybrid Network optical fiber provides the backbone structure while coxl cable provides the connection to the user premises third application is local area network such as 100 base FX Network and thousand base X also uses Fiber Optic Cables there are many advantages associated with fiber optics such as number one they provide high quality transmission of signals at very high

speed number two these are not affected by electromagnetic interference so noise and Distortion is very less l number three they are used for both analog and digital signals they have some disadvantages too such as number one they're expensive number two difficult to install number three maintenance is expensive and difficult and number four they do not allow complete rooting of light signals next is unguided media that is Wireless unguided metor transport electromagnetic waves without using a physical conductor this type of communication is often referred to as a wireless communication signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving

them unguided signals can travel from the source to the destination in several ways such as ground propagation Sky propagation and line of sight propagation figure shows various propagation methods the section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges called bands each regulated by government authorities these bands are rated from very low frequency to extremely high frequency for unguided media transmission and reception are achieved by means of an antenna an antenna can be defined as an electrical conductor or system of conductors used used either for radiating electromagnetic energy or for collecting electromagnetic energy for transmission of a signal electrical energy from the transmitter

is converted into electromagnetic energy by the antenna and radiated into the surrounding environment like atmosphere space or water for reception of a signal electromagnetic energy impinging on the antenna is converted into electrical energy and fed into the receiver in

Satellite telecommunication a down link is the link from a satellite down to one or more ground stations or receivers and an Uplink is a link from a ground station up to a satellite some companies sell Uplink and downlink services to television stations corporations and to other telecommunication carriers a company can specialize in providing uplinks down links or both we can divide Wireless transmission into three Broad groups radio waves microwaves and infrared waves two separate frequencies are used so that

one cannot interfere with the other and full duplex communication is possible and other reason is that the power required to transmit a signal is proportional to the frequency of the signal the more the Power requirement more would be the weight of the system next is radio waves electromagnetic waves ranging in the frequency between 3 khz and 1 GHz are normally called radio waves radio waves are om directional when an antenna transmit radio waves they are propagated in all directions this means that the sending and receiving antennas do not have to be aligned a sending antenna sends waves that can be received by any receiving antenna the OM property has a disadvantage too the radio waves transmitted by one antenna are susceptible to interference

by another antenna that may send signals using the same frequency or band radio waves particularly those waves that propagate in the sky mode can travel long distances this makes radio waves a good candidate for long-distance broadcasting such as AM radio radio waves particularly those of low and medium frequencies can penetrate walls this characteristic can be both an advantage and a disadvantage it is an advantage because for example an AM 22 minutes, 1 second radio can receive signals inside a building it is a disadvantage because we cannot isolate a communication to just inside or outside a building the radio wave band is relatively narrower just under 1 GHz compared to the microwave band now various applications of radio waves the only Direction characteristics of radio

waves make them useful for multicasting in which there is one sander but but many receivers am and FM radio television Maritime radio carless phones and paging are examples of multicasting next is microwaves electromagnetic waves having frequencies between 1 and 3 gahz are called microwaves microwaves are unidirectional when an antenna transmits microwaves they can be narrowly focused this means that the sending and receiving antennas need to be led the unidirectional property has an obvious Advantage a pair of antenas can be aligned without interfering with another pair of aligned atas there are some characteristics of microwave propagation such as number one microwave propagation is line of sight since the towers with the mounted

antenas need to be in direct sight of each other towers that are far apart need to be very tall the curvature of the earth as well as other blocking obstacles do not allow two short Towers to communicate by using microwaves repeaters are often needed for long distance Communications number two very high frequency microwaves cannot penetrate walls this characteristic can be disadvantage if receivers are inside buildings number three the

microwave band is relatively wide almost 299 GHz therefore wider subbands can be assigned and a high data rate is possible and number four use of certain portions of the band requires permissions from authorities now applications of microwaves microwaves due to their unction properties are very useful when unicast communication is needed between

the sender and the receiver they're used in cellular phones satellite networks and Wireless N last but not the least is infrared waves infrared waves with frequencies from 300 GHz to 400 terahz can be used for short range communication infrared waves having high frequencies cannot penetrate walls this advantageous characteristics prevents interference between one system and another a short range communication system in one room cannot be affected by another system in the next room when we use our infrared remote control we do not interfere with the use of the remote by our neighbors however this same characteristic makes infr signals useless for longrange Communications in addition we cannot use infrared waves outside a building because the sun's Rays contain infrared

waves that can interfere with the communication now let us discuss various applications of infrared waves infrared band almost 400 trz has an excellent potential for data transmission such a wide bandwidth can be used to transmit Digital Data with a very high data rate the infrared data Association and Association for sponsoring the use of infrared waves has established standards for using these signals for communications between devices such as keyboards Mouse PCS and printers for example some manufacturers provide a special Port called the irda port that allows a wireless keyboard to communicate with a PC the standard originally defined a dat rate of 75 kilobits per second for a distance up to 8 m the recent standard defines a data rate of 4

MVPs so with this our today's lecture is over today we discussed about various transmission media and their features we also discussed in detail the advantages and disadvantages associated with them I hope you have understood the concepts in the next lecture we will discuss about Wireless networking till then goodbye thank you have a nice day